

Spectral Density Study of the SU(3) Deconfining Phase Transition

Nelsons A. Alves, Bernd Al. Berg and Sergiu Sanielevici

Abstract

We present spectral density reweighting techniques adapted to the analysis of a time series of data with a continuous range of allowed values. In a first application we analyze action and Polyakov line data from a Monte Carlo simulation on $L_t \times L^3$ ($L_t=2,4$) lattices for the SU(3) deconfining phase transition. We calculate partition function zeros, as well as maxima of the specific heat and of the order parameter susceptibility. Details and warnings are given concerning i) autocorrelations in computer time and ii) a reliable extraction of partition function zeros. The finite size scaling analysis of these data leads to precise results for the critical couplings β_c , for the critical exponent ν and for the latent heat $\triangle s$. In both cases ($L_t=2$ and 4), the first order nature of the transition is substantiated.

1 Introduction

We present spectral density reweighting techniques adapted to the analysis of a time series of data with a continuous range of allowed values. In a first application we analyze action and Polyakov line data from a Monte Carlo simulation on $L_t \times L^3$ lattices for the SU(3) deconfining phase transition. We calculate partition function zeros, as well as maxima of the specific heat and of the order parameter susceptibility. Details and warnings are given concerning i) autocorrelations in computer time and ii) a reliable extraction of partition function zeros. The finite size scaling analysis of these data leads to precise results for the critical couplings β_c , for the critical exponent ν and for the latent heat $\triangle s$. In both cases ($L_t=2$ and 4), the first order nature of the transition is substantiated.

2 Method

$L_t \times L^3$ (360) β_c 5.094 ν 5.026 $\triangle s$ 5.1550 $L_t=2$ 0.4060 ν 0.4710 $L_t=4$ 0.4080 β_c 5.090 ν 5.042 $\triangle s$ 5.1220 $L_t=2$ 0.4610 $L_t=4$ 0.4090 β_c 5.090 ν 5.050 $\triangle s$ 5.1080 $L_t=2$ 0.4090 ν 0.4550 $L_t=4$ 0.4110 β_c 5.092 ν 5.064 $\triangle s$ 5.1070 $L_t=2$ 0.4110 ν 0.4540 $L_t=4$ 0.4110 β_c 5.095 ν 5.078 $\triangle s$ 5.1140 $L_t=2$ 0.4130 ν 0.4580

$$L(\theta) = \sum_{i=1}^n \ell(x_i, \theta) \quad (1)$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

math expression && 10,000 && 120,000 (1300) && 5.570 && 5.488 && 5.6710 && 0.4960 &&
0.5560 math expression && 10,000 && 120,000 (2080) && 5.610 && 5.533 && 5.7350 && 0.5120
&& 0.5640 math expression && 10,000 && 120,000 (3000) && 5.640 && 5.566 && 5.7650 &&
0.5220 && 0.5690